

Business Insights Report

Customer Marketing Dataset — Actionable Findings & Recommendations

AI Lab 99 Internship Program 2026 · Module 4 (ML-M04) · Jawad Ahmad

Answers to Key Business Questions

1. Which customers spend the most?

High-income customers without children. These customers fall into the **High-Value Customer** segment (554 of 2,212 customers) and show the highest average spending, driven mainly by Wine and Meat purchases.

2. Which product category generates the highest revenue?

Wine is the top revenue driver (675,296 total units spent), followed by **Meat products** (369,470). Together they represent roughly 74% of total product spending across all six categories.

3. Does income influence spending?

Yes, strongly. Income and Total Spending have a Pearson correlation of **0.79** — one of the strongest relationships in the dataset.

4. Which purchase channel is most popular?

In-store purchases (12,844 total), ahead of web purchases (9,043) and catalog purchases (5,911). Physical retail remains the primary channel.

5. Which campaign performs best?

The **most recent campaign** achieved the highest acceptance rate at 15.1%, clearly outperforming the five historical campaigns (1.4%–7.4%).

6. Which customers respond most frequently?

Higher-income, higher-spending customers are disproportionately represented among repeat campaign responders (the **Campaign Responder** segment, who accepted 2+ campaigns).

7. Which customers are inactive?

Customers with recency above the dataset median (roughly half the customer base) are flagged inactive; the **Inactive Customer** segment (112 customers) represents the most overdue group for re-engagement.

8. Which customers visit the website most?

Customers in the top 10% of monthly website visits — but notably, many of them convert to purchases at a lower rate than expected, indicating a conversion gap rather than a traffic problem.

9. Do customers with children spend differently?

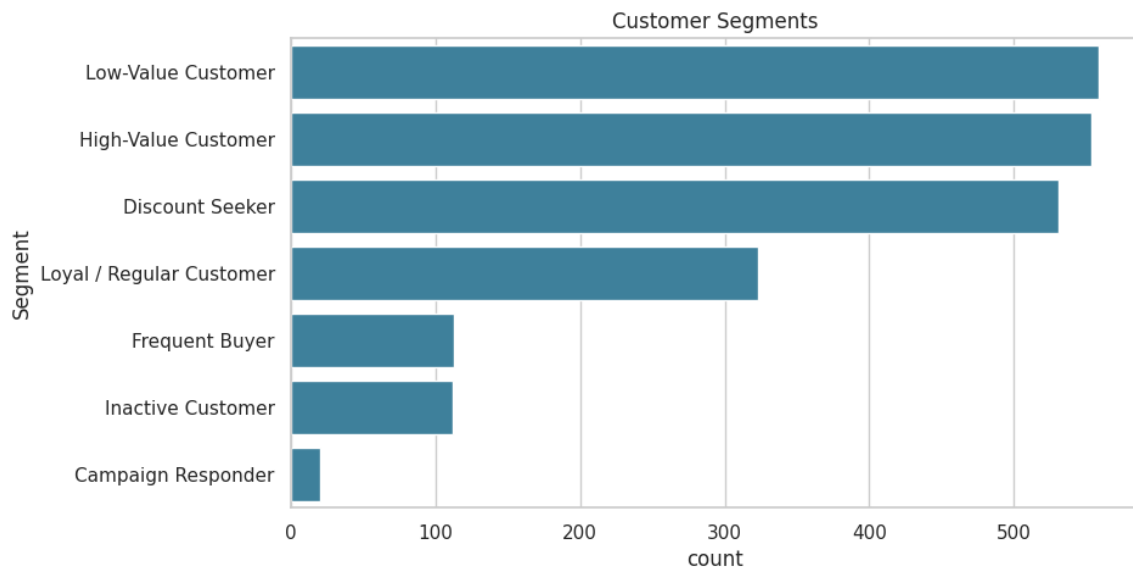
Yes, substantially. Customers **without** children spend an average of PKR/unit 1,104 versus **408** for customers with children — nearly 2.7x more.

10. What characteristics define high-value customers?

Higher income, no or fewer children at home, low reliance on discount purchases, and frequent store and catalog purchases.

11. Which variables have the strongest correlations?

Total Spending correlates most strongly with Wine spend (0.89), Meat spend (0.85), Income (0.79), and Catalog Purchases (0.78); it correlates most negatively with Kidhome (-0.56) and Total Children (-0.50).



Customer segments identified from the analysis.

Recommended Marketing Strategies

- **Launch a targeted re-engagement campaign** for the ~50% of customers with above-median recency, using win-back offers or personalized discounts before they churn.
- **Prioritize Wine and Meat bundles** in promotions and cross-sell campaigns, since these two categories drive the majority of revenue.
- **Target childless, higher-income households** for premium and gold product lines — this group shows the strongest spending potential.
- **Invest in web conversion, not just traffic** — retarget high-visit, low-purchase customers with personalized recommendations or limited-time offers.
- **Model future campaigns after the most recent campaign design**, which more than doubled the acceptance rate of the next-best historical campaign.
- **Build a loyalty program** to reduce dependence on discount-driven purchases and grow the base of full-price, repeat buyers.